12TH BIOLOGY · CHAPTER 20 · FULL CHAPTER MCQS

Biotechnology

51 topic-wise MCQs from the complete chapter with answers and reasons

20.1 – 20.2 GENETIC ENGINEERING AND RECOMBINANT DNA TECHNOLOGY

1. Bread, cheese and beer are products of:

(a) Traditional biotechnology (b) Modern biotechnology
(c) PCR (d) Gene therapy

Answer: (a) Traditional biotechnology — They are made by fermentation.

2. The alteration of an organism's DNA to change its traits is:

(a) Fermentation (b) Hybridization
(c) Genetic engineering (d) Auto-radiography

Answer: (c) Genetic engineering — It is the engine of modern biotechnology.

3. Cutting DNA from one organism and pasting it into another is:

(a) PCR (b) Recombinant DNA technology
(c) Electroporation (d) Screening

Answer: (b) Recombinant DNA technology — It gives a single hybrid DNA molecule.

4. The gene which is to be manipulated is called the:

(a) Vector (b) Primer
(c) Probe (d) Gene of interest

Answer: (d) Gene of interest — It is the first tool of the technology.

5. A gene can be made from its mRNA by using:

(a) Reverse transcriptase (b) DNA ligase
(c) Taq polymerase (d) Restriction endonuclease

Answer: (a) Reverse transcriptase — This is reverse transcription.

6. Restriction endonucleases are also called:

(a) Molecular glue (b) Molecular scissors
(c) Molecular carriers (d) Molecular probes

Answer: (b) Molecular scissors — They cut DNA at specific sequences.

7. The sequence recognized by a restriction endonuclease is the:

(a) Promoter (b) Cos sequence
(c) Restriction site (d) Primer

Answer: (c) Restriction site — Also called the recognition region.

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8. Restriction sites are palindromic sequences of:

(a) 1–3 nucleotides (b) 100–200 nucleotides
(c) 20–30 nucleotides (d) 4–8 nucleotides

Answer: (d) 4–8 nucleotides — Both strands read the same in opposite directions.

9. The single-stranded complementary ends left after cleavage are:

(a) Blunt ends (b) Sticky ends
(c) Primers (d) Probes

Answer: (b) Sticky ends — They let two fragments join.

10. The most commonly used vectors are:

(a) Plasmids of bacteria (b) YACs
(c) Animal viruses (d) Cosmids

Answer: (a) Plasmids of bacteria — They are small bacterial DNA molecules.

11. Plasmids containing the cos sequence of a bacteriophage are:

(a) BACs (b) YACs
(c) Cosmids (d) Probes

Answer: (c) Cosmids — They are a kind of vector.

12. Which of these is NOT a vector?

(a) Phage lambda (b) BAC
(c) Cosmid (d) DNA ligase

Answer: (d) DNA ligase — DNA ligase is an enzyme.

13. DNA ligase forms which linkage?

(a) Hydrogen bond (b) Phosphodiester linkage
(c) Peptide bond (d) Di-sulfide bond

Answer: (b) Phosphodiester linkage — Between two adjacent nucleotides.

14. A vector carrying the gene of interest is called:

(a) Genomic library (b) Probe
(c) Clone (d) Recombinant DNA

Answer: (d) Recombinant DNA — Abbreviated rDNA.

15. Bacteria are an ideal expression system because of their:

(a) Large size (b) Complex genome
(c) Short generation time (d) Cell wall

Answer: (c) Short generation time — And their simple genetic system.

16. In blue-white screening, the desired recombinant colonies are:

(a) White (b) Blue
(c) Red (d) Colourless and dead

Answer: (a) White — The inserted gene breaks the lacZ′ gene.

17. Bacteria that took up a plasmid are selected by growing them on:

(a) Sugar medium (b) Salt medium
(c) Radioactive medium (d) Antibiotic medium

Answer: (d) Antibiotic medium — The plasmid carries an antibiotic resistance gene.

18. The antibiotic named in the textbook for screening is:

(a) Penicillin (b) Ampicillin
(c) Tetracycline (d) Streptomycin

Answer: (b) Ampicillin — Given as the example.

19. A radioactive DNA strand complementary to the gene of interest is a:

(a) Probe (b) Primer
(c) Vector (d) Cosmid

Answer: (a) Probe — It is used in DNA hybridization.

20. Colonies bound to the probe are located by:

(a) Electroporation (b) Microinjection
(c) Auto-radiography (d) Fermentation

Answer: (c) Auto-radiography — Photographic film is used.

20.3 POLYMERASE CHAIN REACTION

21. PCR was developed in 1983 by:

(a) Francis Crick (b) Watson
(c) Khorana (d) Kary Mullis

Answer: (d) Kary Mullis — Nobel Prize in Chemistry, 1993.

22. The primer used in PCR is a sequence of:

(a) 20–30 nucleotides (b) 5–10 nucleotides
(c) 100–200 nucleotides (d) 4–8 nucleotides

Answer: (a) 20–30 nucleotides — It is added in excess.

23. Denaturation in PCR is done at about:

(a) 60°C (b) 72°C
(c) 98°C (d) 37°C

Answer: (c) 98°C — The DNA becomes single-stranded.

24. Annealing of primers occurs at about:

(a) 98°C (b) 60°C
(c) 72°C (d) 20°C

Answer: (b) 60°C — The solution is cooled.

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25. Taq polymerase is isolated from:

(a) Escherichia coli (b) Bacillus thuringiensis
(c) Thermus aquaticus (d) Aspergillus

Answer: (c) Thermus aquaticus — A bacterium of hot springs.

26. Taq polymerase adds new nucleotides to the:

(a) 5′ end of primer (b) Both ends
(c) Middle of template (d) 3′ end of primer

Answer: (d) 3′ end of primer — Just like DNA replication.

27. Each cycle of PCR does what to the number of DNA molecules?

(a) Doubles it (b) Triples it
(c) Adds one (d) Halves it

Answer: (a) Doubles it — 2 → 4 → 8 and so on.

28. After 20 cycles, one DNA fragment gives more than:

(a) One thousand copies (b) One million copies
(c) One hundred copies (d) One billion copies

Answer: (b) One million copies — 2 raised to the power 20.

29. One cycle of PCR is completed in:

(a) One or two minutes (b) One or two hours
(c) One day (d) 20 minutes

Answer: (a) One or two minutes — So PCR is very rapid.

30. DNA fingerprints in criminal investigation are prepared with the help of:

(a) Fermentation (b) PCR
(c) Blue-white screening (d) Microinjection

Answer: (b) PCR — A tiny blood or hair sample is enough.

31. The collection of the total DNA of a single organism is a:

(a) Clone (b) Plasmid
(c) Genomic library (d) Probe

Answer: (c) Genomic library — It is used for sequencing DNA.

20.4 GENETICALLY MODIFIED ORGANISMS

32. GMOs that have received DNA from an unrelated species are:

(a) Clones (b) Mutants
(c) Transgenic organisms (d) Hybrids

Answer: (c) Transgenic organisms — They are a subset of GMOs.

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33. Creating pores in the cell membrane by electric current is:

(a) Microinjection (b) Electroporation
(c) Biolistic transformation (d) Endocytosis

Answer: (b) Electroporation — The pores allow gene intake.

34. A gene gun is used in:

(a) Electroporation (b) Chemical method
(c) Microinjection (d) Biolistic transformation

Answer: (d) Biolistic transformation — The gene is coated with heavy metal ions.

35. Injecting a gene into the nucleus with a fine needle is:

(a) Microinjection (b) Electroporation
(c) Protoplast technique (d) Biolistics

Answer: (a) Microinjection — It is a direct method.

36. Which hormone made by transgenic bacteria treats dwarfism?

(a) Insulin (b) Erythropoietin
(c) Human growth hormone (d) Interferon

Answer: (c) Human growth hormone — Insulin treats diabetes.

37. Erythropoietin from transgenic bacteria is used to treat:

(a) Diabetes (b) Anaemia
(c) Haemophilia (d) Dwarfism

Answer: (b) Anaemia — Clotting factors treat haemophilia.

38. The product that dissolves blood clots is:

(a) Interferon (b) Lysozyme
(c) Chymosin (d) Tissue plasminogen activator

Answer: (d) Tissue plasminogen activator — Made by transgenic bacteria.

39. The enzyme that clots milk protein for cheese making is:

(a) Chymosin (b) Alpha-amylase
(c) Pectin-esterase (d) Lysozyme

Answer: (a) Chymosin — A food product of transgenic bacteria.

40. Transgenic goats produce which human enzyme in milk?

(a) Amylase (b) Chymosin
(c) Lysozyme (d) Insulin

Answer: (c) Lysozyme — It fights diarrhoea-causing bacteria.

41. The first human biological drug from a transgenic goat was an:

(a) Antibiotic (b) Anticoagulant
(c) Antibody (d) Analgesic

Answer: (b) Anticoagulant — It reduces blood clots in surgery or childbirth.

42. A single plant cell can form a new plant because it is:

(a) Transgenic (b) Recombinant
(c) Haploid (d) Totipotent

Answer: (d) Totipotent — So transgenic plants are grown from single cells.

43. Insect-resistant plants carry genes of:

(a) Bacillus thuringiensis (b) Escherichia coli
(c) Thermus aquaticus (d) Aspergillus

Answer: (a) Bacillus thuringiensis — A soil bacterium.

44. Glyphosate is a/an:

(a) Insecticide (b) Vitamin
(c) Antibiotic (d) Herbicide

Answer: (d) Herbicide — Many crops are resistant to it.

45. Genes for beta-carotene in golden rice were taken from:

(a) Daffodil (b) Beans
(c) Wild rice (d) Aspergillus

Answer: (a) Daffodil — Beta-carotene is a precursor of vitamin A.

46. About what percentage of the world's cotton and corn is transgenic?

(a) 10% (b) 50%
(c) 90% (d) 25%

Answer: (c) 90% — Also canola, soybean and sugar beet.

20.5 HUMAN INSULIN IN BACTERIA

47. Human insulin consists of how many polypeptide chains?

(a) One (b) Four
(c) Three (d) Two

Answer: (d) Two — Chain A and Chain B.

48. The bacterium used for making human insulin is:

(a) Escherichia coli (b) Thermus aquaticus
(c) Bacillus thuringiensis (d) Rhizobium

Answer: (a) Escherichia coli — Two separate cultures are used.

49. Chain A and Chain B of insulin are linked by:

(a) Peptide bonds (b) Di-sulfide bonds
(c) Hydrogen bonds (d) Phosphodiester bonds

Answer: (b) Di-sulfide bonds — Sulfur-to-sulfur bridges.

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50. Bacteria producing insulin chains are grown in large industrial:

(a) Incubators (b) Centrifuges
(c) Bioreactors (d) Autoclaves

Answer: (c) Bioreactors — This is the fermentation step.

51. The insulin gene is often attached next to the bacterial gene:

(a) Taq (b) Bt
(c) cos (d) lacZ

Answer: (d) lacZ — It helps the bacteria to express the foreign DNA.

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